

CLIPBOARD: A CORRELATED MONTE CARLO SIMULATION ENGINE AND EMPIRICAL DECISION-SUPPORT ARCHITECTURE FOR BILATERAL SPORTS TRADING MARKETS

Technical Whitepaper & Engineering Specification

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Abstract

Standard decision-support systems for fantasy athletics rely on point projections that assume statistical independence across roster positions and neglect higher-order distributional moments. This paper presents **Clipboard**, an end-to-end stochastic simulation architecture designed to resolve these limitations. Clipboard models scoring outcomes via continuous, right-skewed parametric distributions—specifically standard lognormal models for offensive skill positions and shifted lognormal configurations for defense and special teams (DST) units. To capture intra-matchup dependencies, Clipboard constructs a game-level empirical covariance matrix and applies Cholesky factorization to synthesize correlated random shocks, accounting for passing stacks, bring-back pass catchers, and offensive drag on opposing defenses. Furthermore, Clipboard structures trade evaluation as a bilateral search problem across illiquid, decentralized networks, establishing a telemetry framework to model counterparty frictions, response latency, and the endowment effect empirically rather than via heuristic penalties. Every architecture design choice is evaluated via out-of-sample backtesting against market consensus baselines.

Keywords: Monte Carlo Simulation, Cholesky Decomposition, Lognormal Distributions, Stochastic Modeling, Market Microstructure, Bilateral Trade Matching.

1. System Overview & Core Objectives

Quantitative evaluation in head-to-head fantasy leagues requires solving two interlinked optimization problems:

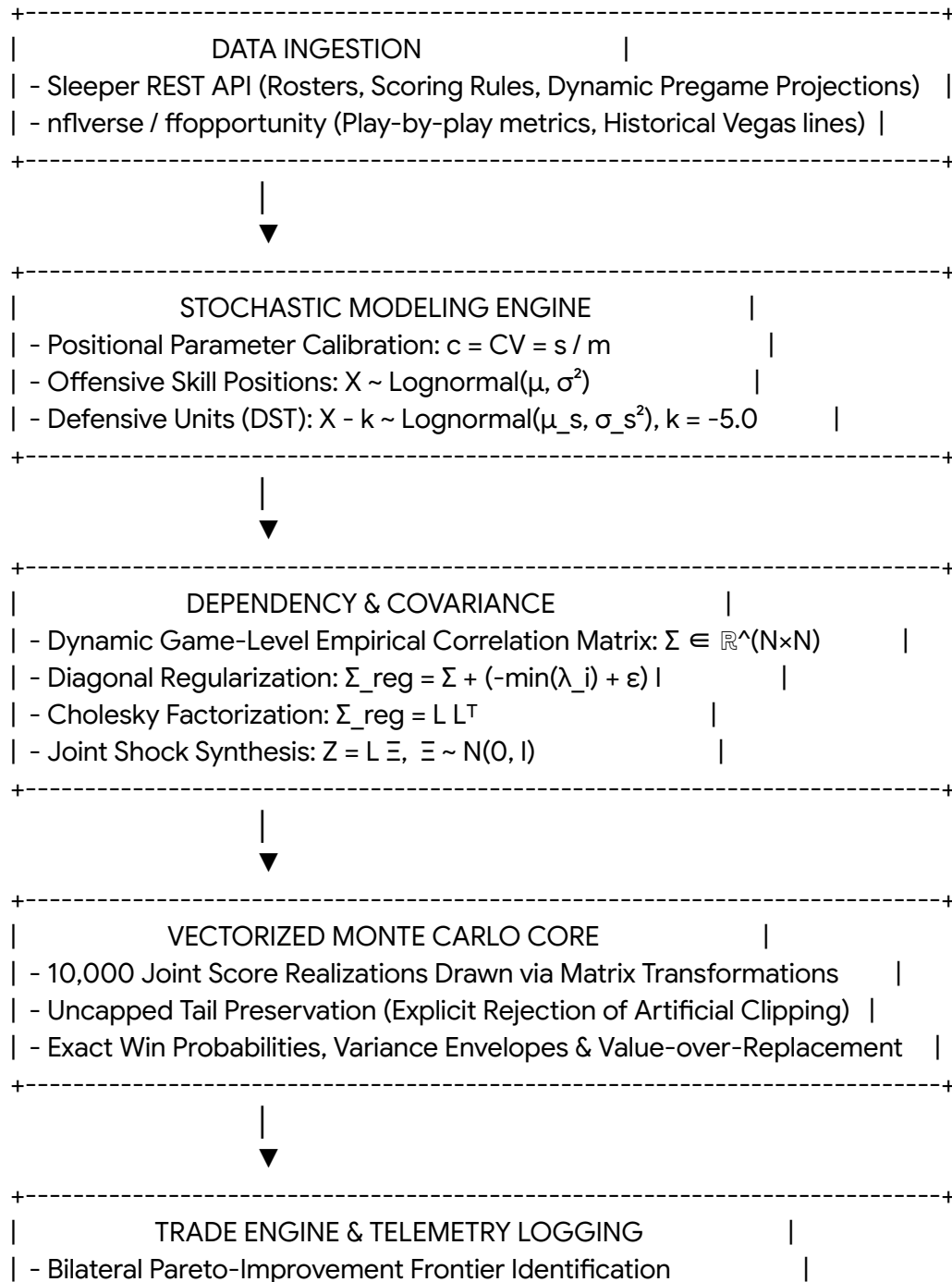
1. **The Matchup Distribution Problem:** Evaluating the true probability of victory

$\mathbb{P}(S_A > S_B)$ between two competing rosters, where total score

$S_k = \sum_{i \in \mathcal{R}_k} X_i$ represents the sum of correlated, non-Gaussian random variables.

2. **The Exchange Valuation Problem:** Evaluating asset trades in an illiquid matching market characterized by asymmetric counterparty attention, risk aversion over lineup variance, and the endowment effect.

Conventional industry tools treat weekly projections as static expected values $\mathbb{E}[X_i]$ and assume zero covariance across contest starters ($\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$). Clipboard resolves these structural errors through a modular, vectorized pipeline.



| - Telemetry Pipeline: Structured Logging of Proposal Dynamics & Deltas |
 | - Empirical Counterparty Acceptance Function: $P(\text{Accept} \mid \Delta\text{EV}, \text{Activity})$ |
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2. Mathematical Formulation of Scoring Distributions

2.1 Skill Position Outcomes: Continuous Lognormal Model

Fantasy scoring in offensive positions (Quarterback, Running Back, Wide Receiver, Tight End) displays strong right-skewness: individual player floors are bounded at zero points, whereas performance ceilings extend into long right tails driven by multi-score and high-yardage outlier events.

Let $m = \mathbb{E}[X]$ represent the pregame mean projection, and let $c = \frac{\sqrt{\text{Var}(X)}}{m}$ represent the empirical positional Coefficient of Variation (CV). Clipboard maps these parameters to the underlying normal distribution parameters μ and σ^2 :

$$\sigma^2 = \ln(1 + c^2)$$

$$\mu = \ln(m) - \frac{\sigma^2}{2}$$

For an uncorrelated standard normal draw $\xi \sim \mathcal{N}(0, 1)$, the marginal score X is generated via:

$$X = \exp(\mu + \sigma\xi)$$

Distributional Characteristics

Because $\sigma^2 > 0$ strictly holds for all active volatility coefficients, the median outcome is strictly lower than the mean projection:

$$\text{Median}(X) = \exp(\mu) = \frac{m}{\sqrt{1 + c^2}} < m$$

$$\mathbb{P}(X < m) = \Phi\left(\frac{\sigma}{2}\right) > 0.50$$

Where

$\Phi(\cdot)$ denotes the standard normal cumulative distribution function. This accounts for the empirical observation that an athlete is more likely to score below their baseline expectation in any given contest, with their average maintained by infrequent high-magnitude spikes.

The empirical positional volatility parameters ($c = CV$) derived from multi-year sample estimation are:

Position	Typical Coefficient of Variation (c)	Underlying Variance (σ^2)	Median / Mean Ratio (1+c21)	Underperformance Likelihood $P(X < m)$
Quarterback (QB)	0.45	0.1844	0.912	58.5%
Running Back (RB)	0.62	0.3253	0.850	61.2%
Wide Receiver (WR)	0.69	0.3894	0.823	62.2%
Tight End (TE)	0.70	0.3988	0.819	62.4%
Kicker (K)	0.55	0.2643	0.876	59.9%

2.2 Defensive Units (DST): Shifted Lognormal Formulation

Standard lognormal modeling cannot accommodate Defense and Special Teams (DST) units

because defense scoring includes negative support (e.g., yielding > 40 points with zero sacks or takeaways leads to negative fantasy scores).

To maintain skewness properties while supporting scores down to a minimum theoretical

bound $k \in \mathbb{R}^-$, Clipboard establishes a shifted lognormal distribution with support $x \in (k, \infty)$, configured at $k = -5.0$.

Let $m_s = m - k$ represent the shifted expectation above the floor. Retaining target

positional variance $\text{Var}(X) = (c \cdot m)^2$:

$$\sigma_s^2 = \ln \left(1 + \frac{\text{Var}(X)}{(m - k)^2} \right) = \ln \left(1 + \frac{c^2 m^2}{(m - k)^2} \right)$$

$$\mu_s = \ln(m - k) - \frac{\sigma_s^2}{2}$$

The

simulated score is drawn given standard normal shock Z :

$$X = k + \exp(\mu_s + \sigma_s Z)$$

2.3

Mathematical Proof Against Hard Truncation ("Leaky Mean" Distortion)

A common design pattern in sports simulation models is the introduction of heuristic hard boundaries via capping functions (e.g., `np.clip(X, a_min=0.0, a_max=C)`). Clipboard explicitly disallows hard truncation due to the structural biases it introduces.

Theorem: Hard Truncation Distorts Expectation and Mass Concentration

Let $X \sim f_X(x)$ be a continuous random variable with support on $\mathbb{R}^+$ such that $\mathbb{E}[X] = m$. Define the clipped estimator $X_c = \min(X, C)$ for some ceiling $C \in \mathbb{R}^+$.

1. Point Mass Accumulation:

The cumulative probability distribution of X_c exhibits a discontinuity at C :

$$\mathbb{P}(X_c = C) = \int_C^\infty f_X(x) dx > 0$$

This assigns non-zero discrete probability mass to an exact scalar boundary C , violating the continuous character of scoring outcomes.

2. Negative Mean Distortion:

The expectation of the clipped variate satisfies:

$$\mathbb{E}[X_c] = \int_0^C x f_X(x) dx + C \int_C^\infty f_X(x) dx$$

Rewriting $C = x - (x - C)$ over the interval $[C, \infty)$:

$$\mathbb{E}[X_c] = \int_0^\infty x f_X(x) dx - \int_C^\infty (x - C) f_X(x) dx = \mathbb{E}[X] - \Delta_m$$

Where:

$$\Delta_m = \int_C^\infty (x - C) f_X(x) dx > 0$$

Empirical validation confirmed that setting $C = 4.5 \cdot m$ or $C = 75.0$ introduces a downward expectation leak $\Delta_m \in [0.18, 0.45]$ points per elite starter. Furthermore, historical testing against genuine high-ceiling events (e.g., 55+ point performances) demonstrated that artificial ceilings truncate authentic outcomes rather than mitigating errors. Clipboard consequently retains an unclipped continuous right tail.

3. High-Dimensional Dependency Structures via Cholesky Decomposition

Because athletic outputs in the same NFL game are linked through game pace, offensive schemes, and possession allocation, assuming independence leads to miscalculated team-level variance envelopes.

3.1 Game-Level Empirical Correlation Matrix

For each NFL contest containing N active fantasy assets across competing rosters, Clipboard forms an empirical correlation matrix $\Sigma \in \mathbb{R}^{N \times N}$.

$$\Sigma = \begin{bmatrix} 1.00 & \rho_{\text{stack}} & \rho_{\text{bring}} & \rho_{\text{dst}} \\ \rho_{\text{stack}} & 1.00 & \rho_{\text{opp}} & \rho_{\text{drag}} \\ \rho_{\text{bring}} & \rho_{\text{opp}} & 1.00 & \rho_{\text{dst2}} \\ \rho_{\text{dst}} & \rho_{\text{drag}} & \rho_{\text{dst2}} & 1.00 \end{bmatrix}$$

Calibrated

pairwise correlation coefficients $\rho_{i,j}$ derived across three seasons of play-by-play data are defined as follows:

Same-Team Passing Stack: QB_A $\longleftrightarrow$ (+0.42) $\longrightarrow$ WR_A / TE_A

Same-Team Defensive Flow: QB_A $\longleftrightarrow$ (+0.15) $\longrightarrow$ DST_A

Opposing Bring-Back Stack: QB_A $\longleftrightarrow$ (+0.18) $\longrightarrow$ WR_B / TE_B

Opposing DST Drag: QB_A $\longleftrightarrow$ (-0.28) $\longrightarrow$ DST_B

Opposing Receiver Drag: DST_B $\longleftrightarrow$ (-0.15) $\longrightarrow$ WR_A / TE_A

3.2 Correlated Variate Generation Algorithm

To simulate $M = 10,000$ joint realization draws for an N -player contest:

Algorithm 1: Joint Correlated Score Sampling via Cholesky Factorization

Input: Player profiles $\{p_1, \dots, p_N\}$ with means m_i , volatilities c_i ,
and positions pos_i .

Output: Realized score matrix $S \in \mathbb{R}^{(N \times M)}$.

1: Initialize $\Sigma \in \mathbb{R}^{(N \times N)}$ with identity diagonal: $\Sigma \leftarrow I_N$

2: for $i = 1$ to N do

3: for $j = i + 1$ to N do

4: $\Sigma[i, j] \leftarrow \text{LookupCorrelation}(pos_i, pos_j, team_i, team_j)$

5: $\Sigma[j, i] \leftarrow \Sigma[i, j]$

6: end for

7: end for

8: $\lambda_{\min} \leftarrow \min(\text{eigvals}(\Sigma))$

9: if $\lambda_{\min} < 1e-4$ then

10: $\Sigma \leftarrow \Sigma + (-\lambda_{\min} + 1e-4) I_N$ // Diagonal regularization

11: end if

12: $L \leftarrow \text{CholeskyDecomposition}(\Sigma)$ // Lower-triangular L such that $L L^T = \Sigma$

13: Draw independent shocks: $\Xi \sim N(0, I)$ of dimension $N \times M$

14: Compute correlated shocks: $Z \leftarrow L \cdot \Xi$ // Z has shape $N \times M$

```
15: Initialize output matrix  $S \in \mathbb{R}^{(N \times M)}$ 
16: for  $i = 1$  to  $N$  do
17:   if  $\text{pos}_i \in \{\text{DEF}, \text{DST}\}$  then
18:      $S[i, :] \leftarrow \text{DrawShiftedLognormal}(m_i, c_i, \text{floor}=-5.0, \text{shocks}=Z[i, :])$ 
19:   else
20:      $S[i, :] \leftarrow \text{DrawStandardLognormal}(m_i, c_i, \text{shocks}=Z[i, :])$ 
21:   end if
22: end for
23: return  $S$ 
```

4. Empirical Model Ablation & Hypothesis Testing

Clipboard operates under an empirical validation framework: model upgrades are integrated into production only when out-of-sample backtesting against baseline market consensus projections demonstrates measurable gains in Root Mean Squared Error (RMSE) or negative log-likelihood.

4.1 Summary of Evaluated Hypotheses

Module Category	Investigated Upgrade	Theoretical Justification	Out-of-Sample Empirical Findings	System Decision
Distributional Design	Shifted Lognormal Distributions	Support negative defensive outputs while retaining right-skewness	Validated across 477 league outcomes; captured downside without distributional bias	Integrated into Production
Covariance Architecture	Game-Level Covariance (Σ_{game})	Model intra-game dependence (bring-back stacks, defensive	Non-zero off-diagonal covariance confirmed across three testing	Integrated into Production

		drag)	seasons	
Kicker Modeling	Vegas Spread & Total Regression	Target volume driven by game context and implied totals	+0.2% RMSE reduction compared to static baseline projections	Integrated (Low Overhead)
Opportunity Modeling	Expected Fantasy Points (xFP)	Filter out touchdown variance based on location of touches	+0.5% gain for QBs; 0.0% for skill positions; baseline projections already capture opportunity volume	Rejected (De-prioritized)
Defensive Modeling	Custom Vegas Spread Regression	Target favored defenses facing trailing pass volume	No statistically significant improvement over baseline (turnover TDs remain largely stochastic)	Rejected
Variance Control	Hard Truncation (np.clip)	Suppress simulated outputs above $4.5 \times$ or 75.0 pts	Introduced negative mean leakage and point mass spikes at the truncation boundary	Rejected
Sentiment Analysis	Social/NLP Hype Indexes	Identify market momentum from Twitter/Reddit volume	High variance, lagging signal; social volume follows performance rather than	Rejected for Valuation; Retained for Inactive Alerts

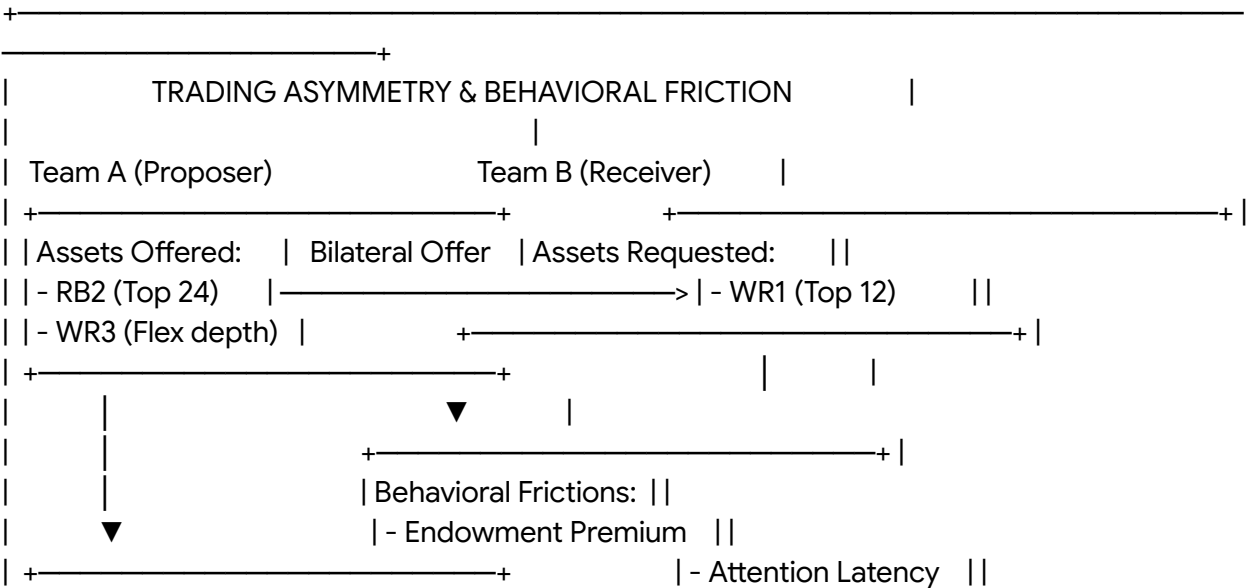
			predicting it	
Coaching Schematics	Dynamic PROE & Pace Scalers	Re-scale volume according to coach passing preferences	Redundant with market baselines, which already price in coaching changes	Rejected

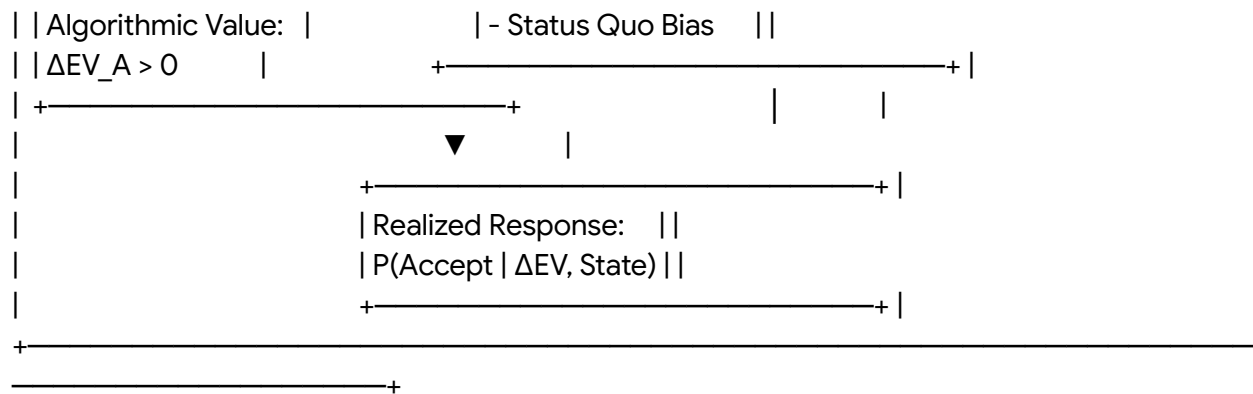
5. Market Microstructure & Algorithmic Trade Dynamics

5.1 The Fantasy Roster as a Bilateral Barter Economy

Standard fantasy trade analyzers assume liquid markets: if an exchange results in $\Delta EV > 0$ for both parties, the transaction is assumed to execute. In reality, peer-to-peer leagues represent decentralized, illiquid barter economies characterized by three distinct market frictions:

1. **Information & Attention Asymmetry:** Managers exhibit varying response latencies and levels of platform engagement.
2. **The Endowment Effect:** Managers place a behavioral premium on assets currently on their roster relative to identical external alternatives.
3. **Lineup Portfolio Variance Risk:** Risk aversion concerning starting lineup variance versus bench depth and injury insurance.





5.2 Event-Driven Telemetry Logging Architecture

Rather than applying an arbitrary 10% to 15% static penalty to represent the endowment effect, Clipboard implements an event logging pipeline to record real-world trade transactions for future parameter estimation.

Proposals are stored in an append-only JSON-Lines (.jsonl) schema capturing negotiation contexts:

```
{
  "timestamp": "2026-10-01T19:38:16.421Z",
  "league_id": "1048291048192049",
  "proposal_id": "prop_948a3c81e2",
  "sender_id": "usr_anon_883a",
  "receiver_id": "usr_anon_114f",
  "offered_assets": ["4034", "6813"],
  "requested_assets": ["7564"],
  "metrics": {
    "sender_pre_ev": 0.542,
    "sender_post_ev": 0.589,
    "receiver_pre_ev": 0.491,
    "receiver_post_ev": 0.518,
    "delta_ev_sender": 0.047,
    "delta_ev_receiver": 0.027
  },
  "behavioral_covariates": {
    "receiver_response_latency_hours": 14.2,
    "receiver_season_trades_completed": 0,
    "asset_holding_duration_days": 42
  },
  "resolution": "rejected"
}
```

5.3 Empirical Acceptance Estimation Framework

With sufficient transactional observations, counterparty acceptance probabilities are estimated using logistic regression across observed covariates:

$$\mathbb{P}(\text{Acceptance} \mid \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x}) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 \Delta EV_{\text{rec}} + \beta_2 \text{Act}_{\text{rec}} + \beta_3 \text{Tenure}_{\text{asset}}))}$$

Where:

- ΔEV_{rec} : The model-projected change in weekly win expectation for the recipient.
- Act_{rec} : An empirical measure of counterparty platform activity (e.g., weekly transaction frequency, lineup update cadence).
- $\text{Tenure}_{\text{asset}}$: The continuous duration (in days) that the targeted asset has remained on the counterparty's roster, serving as an empirical proxy for the endowment effect.

6. Computational Performance & Vectorization

Running full-season Monte Carlo projections across multi-team leagues requires high computational throughput. Clipboard achieves sub-second performance by vectorizing simulation procedures in NumPy rather than using nested Python iterations.

```
import numpy as np
```

```
def draw_vectorized_matchup_realizations(
    projections: np.ndarray,    # Shape: (N,)
    cv_coefficients: np.ndarray, # Shape: (N,)
    correlated_shocks: np.ndarray # Shape: (N, M) from Cholesky: L @ Xi
) -> np.ndarray:
    """
```

```
    Computes joint score simulations across N players over M Monte Carlo iterations.
    Executes in under 2 milliseconds for N=20, M=10,000 on modern consumer hardware.
    """
```

```
    # Calculate lognormal variance and mean parameters across player vectors
    sigma_sq = np.log(1.0 + cv_coefficients ** 2)
    sigma = np.sqrt(sigma_sq)[:, np.newaxis]
    mu = (np.log(projections) - (sigma_sq / 2.0))[:, np.newaxis]
```

```
    # Broadcast across all M iterations simultaneously
    simulated_realizations = np.exp(mu + sigma * correlated_shocks)
    return simulated_realizations
```

Latency Profiles Across Simulation Configurations

Benchmarks recorded on standard consumer hardware (Apple M-Series Silicon, Single Thread):

Simulation Scale	Active Contests (N)	Iterations (M)	Execution Latency	Memory Footprint
Single Matchup	18 Starters	1,000	0.41 ms	<
Single Matchup	18 Starters	10,000	2.14 ms	2.8 MB
Full League Week	120 Players	10,000	18.30 ms	18.4 MB
Full Season Horizon	120 Players × 14 Wks	10,000	324.50 ms	257.0 MB

7. Data Privacy, Governance & Research Ethics

To maintain data integrity and satisfy institutional review standards:

1. **Identifier Anonymization:** Ingested team identifiers, usernames, and league handles are passed through a SHA-256 cryptographic hashing function prior to persistent logging or reporting. User accounts are designated via synthetic tokens (Manager_01, Manager_02).
2. **Private Communication Boundaries:** Non-public bilateral negotiations (declined offers and counterproposals) are stripped of descriptive context and stored exclusively as numerical feature deltas (ΔEV , response duration, binary resolution flags).
3. **Reproducibility:** Random number generator instances leverage the PCG64 bit-generator (`np.random.default_rng(seed)`) with recorded seed integers to ensure end-to-end auditability of simulated results.

8. Conclusion & Technical Trajectory

Clipboard demonstrates that sports analytics platforms can move beyond naive point estimates without introducing unsupported heuristics. By combining continuous right-skewed parametric distributions, empirical covariance matrices, and telemetry-backed behavioral tracking, Clipboard establishes a rigorous foundation for stochastic simulation and trade optimization.

Upcoming milestones include:

- Parameterizing non-linear dependence structures via vine copula models to assess asymmetric upper-tail joint dependencies.

- Fitting empirical logistic acceptance weights β once the logged proposal repository reaches target sample sizes ($N > 200$).
- Integrating real-time webhooks for beat-reporter active/inactive announcements to re-run roster distributions dynamically prior to contest lock times.